

Gal Moyal & Eden Furman

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1. BACKGROUND & THE NEED



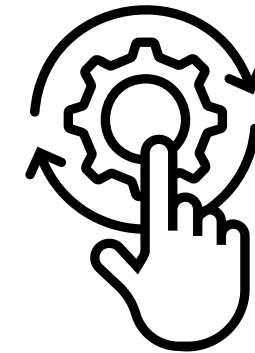
The Academic Problem: Academic timetabling is a complex NP-hard optimization problem. Each institution has its own unique rules, structures and constraints



The Current Manual Workflow: In the Software Engineering Department at Braude College, the scheduling process is performed manually



The Complexity of Constraints: A valid schedule must satisfy many requirements. They are divided into **Hard Constraints** (must be met) and **Soft Constraints** (improve schedule quality)

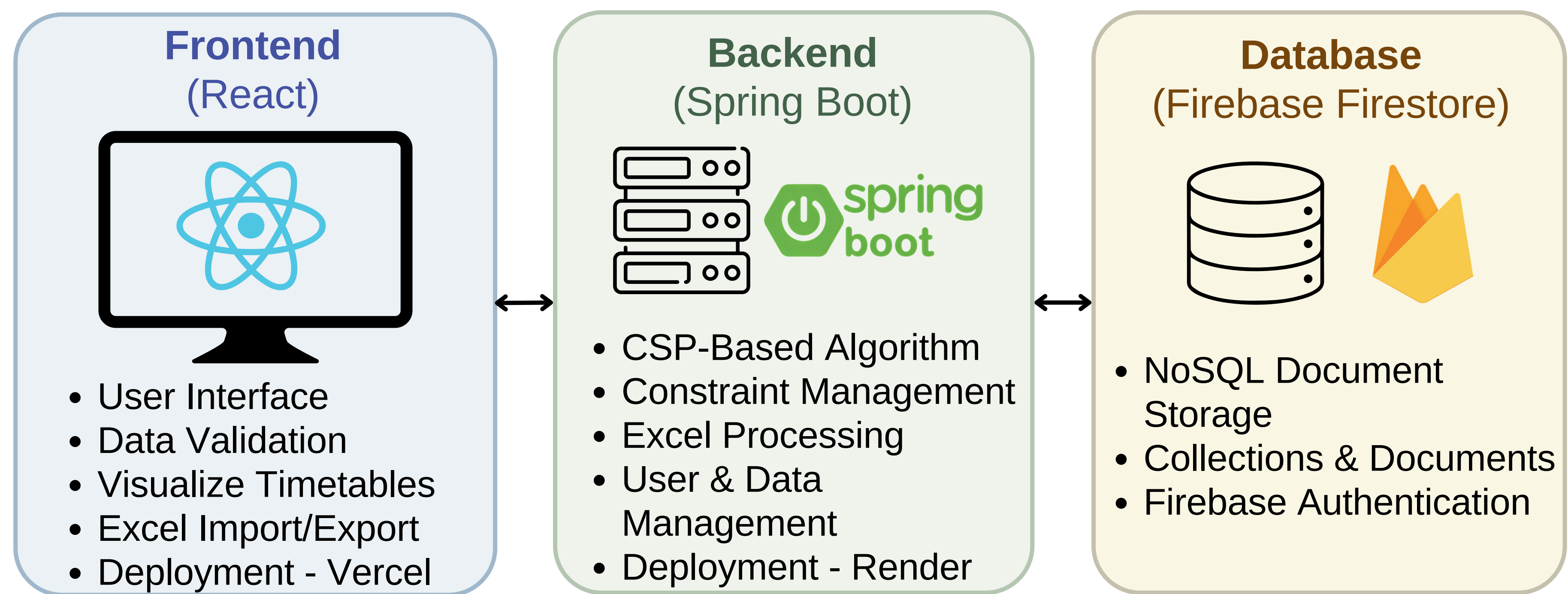


The Need for Replacement: An automated and efficient scheduling system is needed

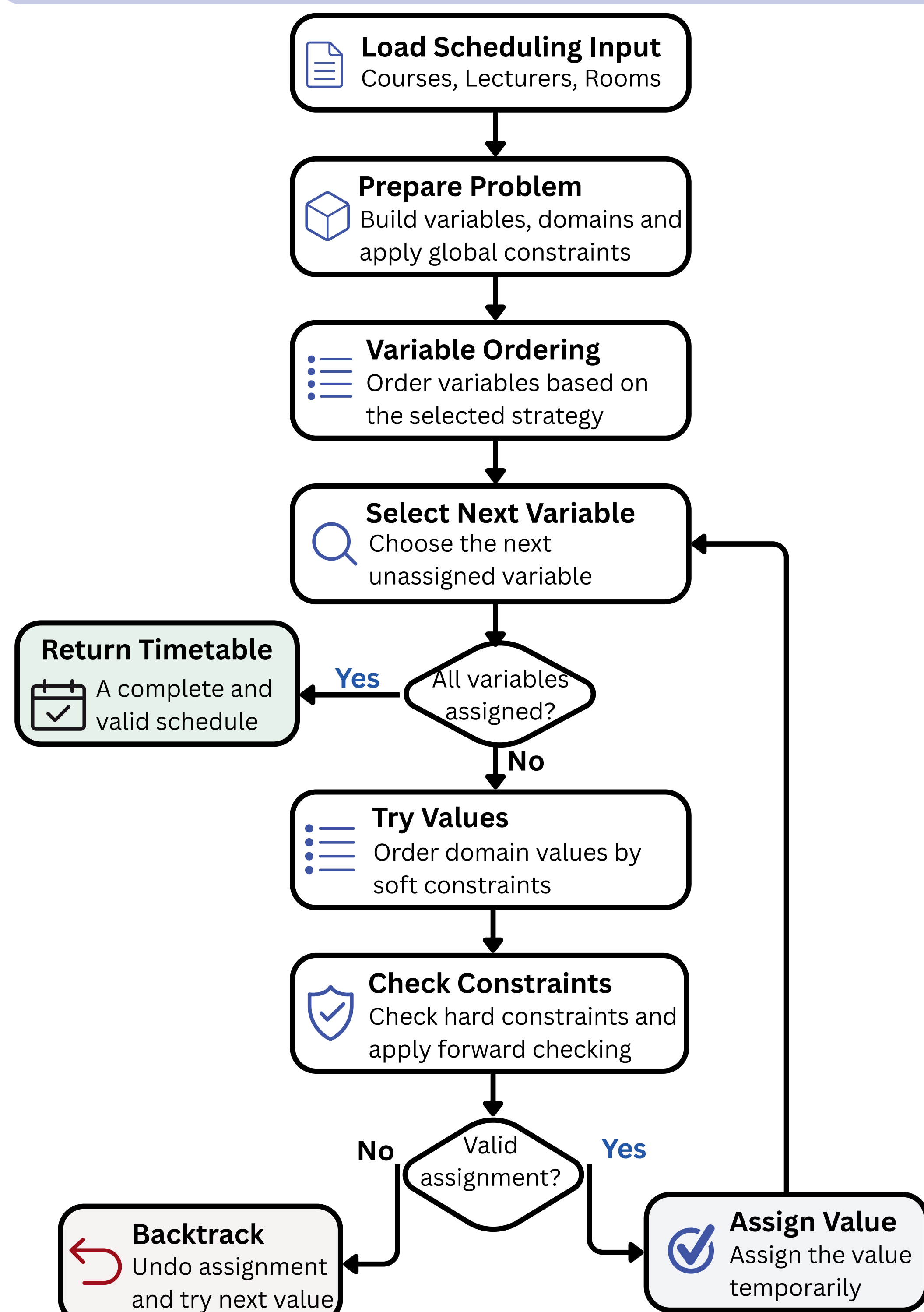
2. CORE SYSTEM REQUIREMENTS

- User-Friendly Interface:** Intuitive platform designed for department coordinators
- Data Management:** Easy handling of data: courses, lecturers, classrooms
- Smart Generation:** Automatic scheduling adhering strictly to all hard constraints
- Dynamic Flexibility:** User control over Soft Constraint weights to match changing semester preferences

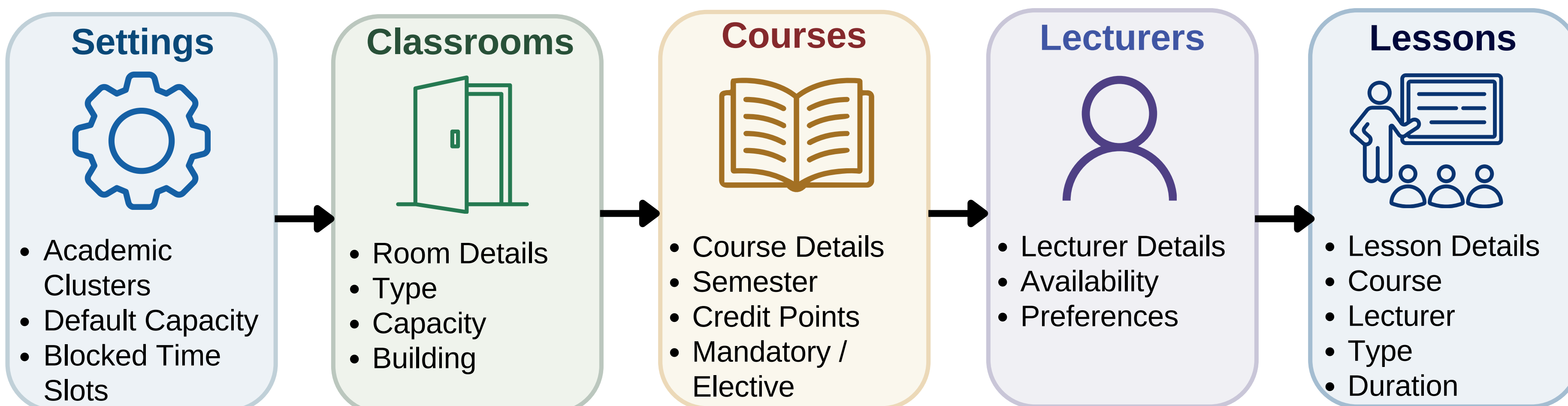
3. SYSTEM ARCHITECTURE



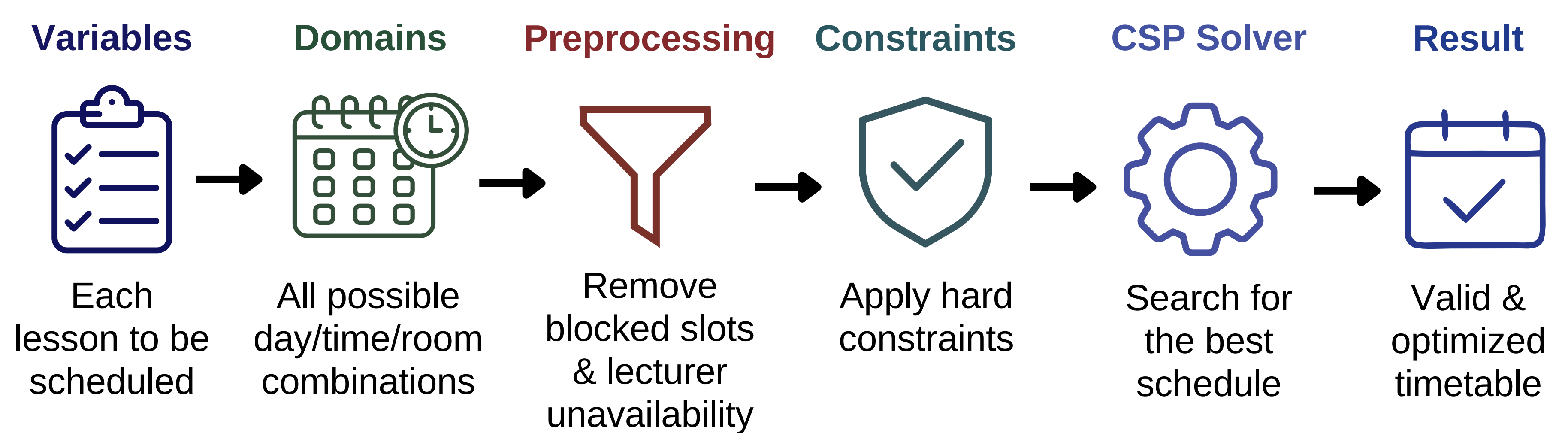
6. ALGORITHM EXECUTION FLOW



4. DATA PREPARATION WORKFLOW



5. CSP SCHEDULING MODEL



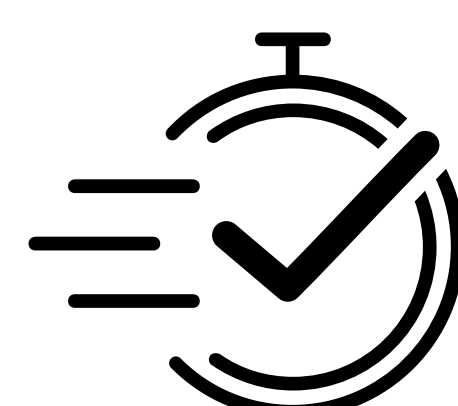
7. WEIGHTED COST FUNCTION

$$\text{Raw Penalty } P_i \div \text{Max Possible Penalty } \max P_i = \text{Normalized Penalty } N_i \times \text{Weight } W_i = \text{Weighted Penalty } N_i \times W_i$$

$$\text{Total Assignment Cost: } \sum N_i \times W_i$$

Lower cost = Better assignment

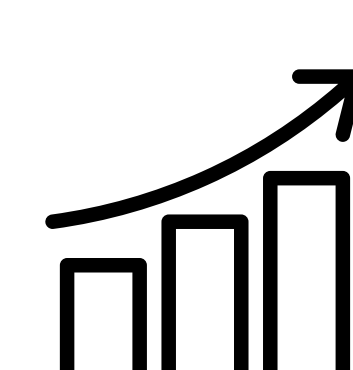
8. KEY BENEFITS



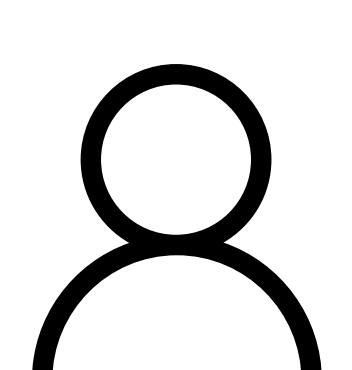
Save Time
Automated generation in minutes instead of days



Constraint Compliance
All hard constraints are strictly enforced



Better Quality
Optimized schedule with soft constraints

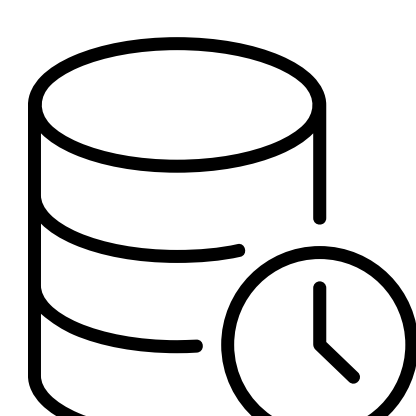


User Friendly
Easy data management & flexible preferences

9. PROJECT INSIGHTS



Constraints Identification
Defined and refined scheduling constraints



Free-Tier Limitations
Implemented caching & improved efficiency



Client Collaboration
Continuous feedback shaped the final system

Link to GitHub:
<https://github.com/GalMoyal111/course-scheduling-system.git>